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Sep 16

Polyimide (PAI) materials are predominantly aromatic molecules of high thermal stability. They are usually melt-processed, but are infusible in their fully finished form after post-curing.

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reinforced thermoplastic, but are extremely expensive.
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Properties approaching those of Polyimide, yet melt processable. Highest strength of any unreinforced thermoplastic. Good wear resistance. Low smoke emission. Good radiation resistance.

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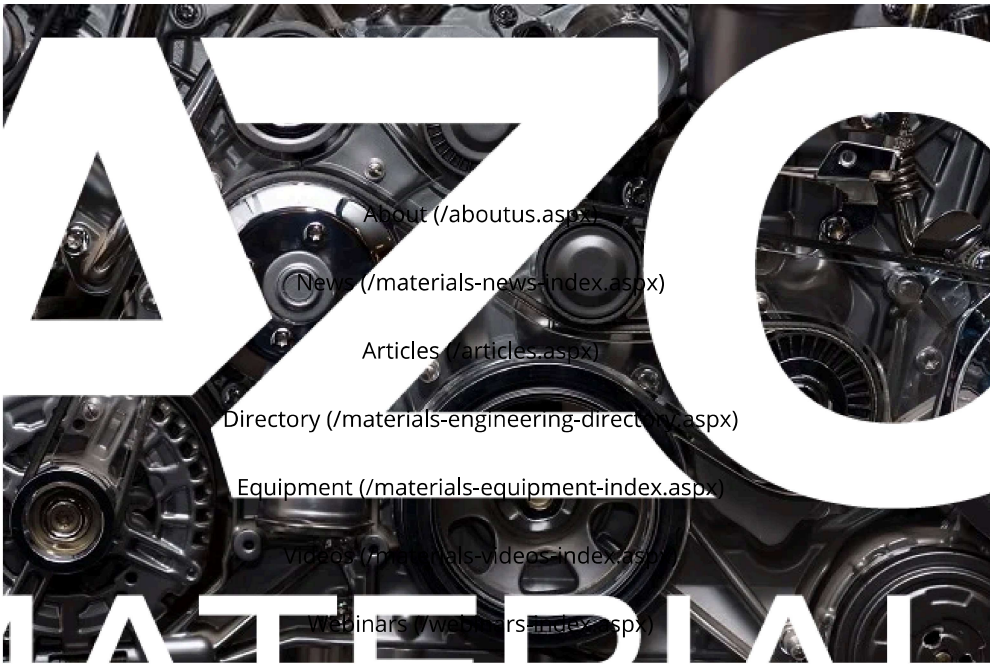
Applications of PAI include valves, bearings, electrical connectors, gears and general mechanical components.

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Typical Properties

Property

Density (g/cm³)

Surface Hardness

Tensile Strength (MPa)

Flexural Modulus (GPa)

Notched Izod (kJ/m)

Linear Expansion (°C x 10⁻⁵)

Elongation at Break (%)

Strain at Yield (%)

Max. Operating Temp. (°C)

Water Absorption (%)

Oxygen Index (%)

Flammability UL94

Volume Resistivity (log ohm.cm)

Dielectric Strength (MV/m)

Dissipation Factor 1 kHz

Dielectric Constant 1 kHz

HDT @ 0.45 MPa (°C)

HDT @ 1.80 MPa (°C)

Material. Drying hrs @ (°C)

Melting Temp. Range (°C)

Mould Shrinkage (%)

Mould Temp. Range (°C)

Source : Abstracted from Plascams

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RM109

185

4.58

0.13

3.6

12

8

210

0.28

43

V0

17

23

0.001

3.5

260

274

16 @ 150

315 - 360

0.7

180 - 260

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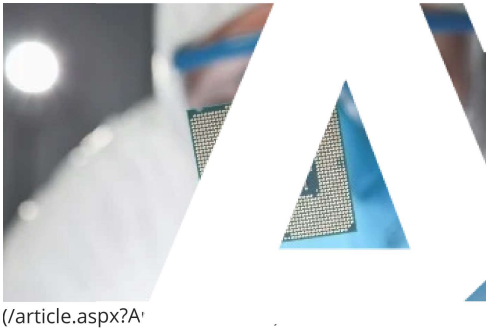
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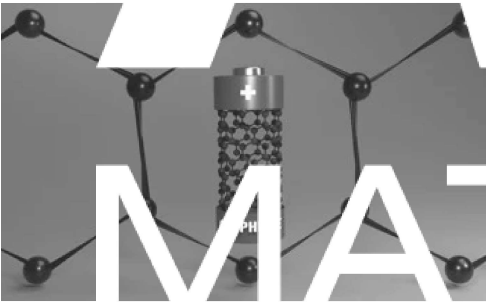
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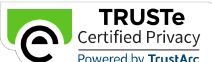
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